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	The principal handles that we will use to influence solution behavior and crystallization are surfactants, electrolytes and hydrophilic polymers. The specific application to membrane protein crystallization will rest on characterization of protein and micellar interactions by scattering and chromatographic methods, and tuning of these interactions to guide crystallization. To reach our objective we will pursue the following specific steps: 1) Production and purification of the membrane protein of interest; 2) Measurement of interactions between protein-surfactant complexes; 3) Growth of crystals by tuning of solution conditions based on interaction measurements;		

Task Description:	4) Characterization of diffraction quality of crystals and determination of protein structure. We work with three membrane proteins, two mammalian and one bacterial. The systems are also amenable to mutation to allow investigation of the role of specific structural features in crystallization. Thus our approach is one that we expect to yield not just suitable crystals in at least a subset of the systems to be examined, but more importantly also systematic approaches for obtaining and characterizing diffraction-quality crystals of membrane proteins that can be applied to a wide variety of systems. Identification and control of the physical chemical aspects can also contribute directly to the framework needed to understand the unique features of membrane protein crystallization in microgravity. Our objective is to measure, manipulate and model the interactions among membrane proteins and their complexes with detergents, with the over-arching goal of controlling crystallization by rational manipulation of these interactions. The Protein Data Bank now contains more than 20000 structures, with more than 3000 being deposited annually, and the rate of growth is expected to expand dramatically as structural genomics efforts reach fruition. In contrast to this larger picture, only a few dozen structures of integral membrane proteins have been deposited. As a result, the structural biology of membrane proteins is much less well understood than that of soluble proteins, which is a serious shortcoming not just because of the number of membrane proteins that occur in nature, but also because of their critical role as receptors, ion channels, and membrane-bound enzymes, which makes them major targets for drug development. The availability of more structures of membrane proteins would allow rational drug design in addition to the alternative routes to discovery that must play a more important role in the absence of 3D structures. Thus there is a large potential incremental benefit of additional membrane protein structures to both the structural biology and drug design communities, and an especially large one of a fundamental understanding of membrane protein crystallization that can lead to generally applicable procedures, such as will be sought in our proposed research.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the interactions among membrane proteins and their complexes with detergents, will allow enhanced protein purification and can lead to new crystal structures of materials useful in pharaceutical applications.
Task Progress:	Membrane Protein Activity and Recovery: A major challenge in studying membrane proteins is identifying optimal conditions for their stabilization outside of the bilayer. Experimental results for bacteriorhodopsin, E. coli diacylglycerol kinase, human adeonsine A2a and A3 receptor using 50 surfactants common to membrane protein purification reveals a relationship between the hydrophile-lipophile balance (HLB) of a given surfactant and the corresponding activity of the membrane protein, with a differing window of optimal HLB values for each. Comparing these results against a detailed analysis of integral membrane proteins of known structure, a strong correlation is observed between the accessible transmembrane surface area of each membrane protein and the HLB of the surfactant used to crystallize or purify it, demonstrating that HLB is an appropriate quantity to correlate with membrane protein activity, and provides a quantitative link between surfactant properties and membrane protein function. This link can be understood in terms of the molecular packing parameter, which is inversely related to HLB, such that the observed trends with HLB and membrane protein activity are a consequence of a surfactant's ability to pack effectively around the transmembrane region of a membrane protein. Therefore, the observed correlation with membrane protein activity, HLB and accessible surface area makes sense in terms of the shape and size of a membrane protein. In order to estimate surfactants within an optimal window of HLB values, however, the accessible transmembrane surface area must be known. Therefore, we examined several properties of membrane proteins, which led to a strong correlation between accessible surface area (as well as molecular volume) and molecular weight. The linear relationship between accessible surface area and molecular weight is somewhat surprising, however, considering previous work with soluble, globular proteins found considerably different results. However, when considered in the context of the HLB - accessible surface area correlation, one might expect that membrane proteins are significantly elongated in the direction perpendicular to the plane of the membrane, and therefore the non-spherical packing parameter values reflect a cylindrical, rather than spherical geometry, which would lead to a linear correlation with molecular weight. Overall, however, these results hold great promise for improving and developing new methods for membrane protein research. Membrane Protein Crystallization: The interactions leading to crystallization of the integral membrane protein bacteriorhodopsin solubilized in n-octyl-ß-D-glucoside were investigated. Osmotic second virial coefficients were measured by self-interaction chromatography in the presence of salts such as sodium malonate, sodium formate and ammonium sulfate as well as a wide range of additives and polyethylene glycol (PEG). Our results demonstrate that the surfactant plays a key role in protein-detergent complex (PDC) interactions leading to crystallization. Dynamic light scattering and tensiometry measurements show that the interaction trends are strongly influenced by micelle structure and surfactant phase behavior, both of which are sensitive to salt and surfactant concentration. We have shown the dual role of the surfactant in enhancing PDC attraction near the cloud point temperature as well as indirect action through excluded volume effects on PDC interactions. The latter may be important in preventing PDC interactions and in reducing the range of precipitant concentrations conducive to weak attraction. Minimizing this effect can be accomplished by reducing surfactant concentration to near CMC values. Additives such as heptane-1,2,3-triol and hexane-1,6-diol, which raise the 'cloud point' temperature of surfactant solutions, are likely affecting the free micelles in solution rather than PDCs to promote crystallization. Addition of either to C8ßG1 significantly raises the CMC and causes the micelle size and polydispersity to become independent of ionic strength. However, a corresponding shift in attractive PDC interactions with increasing 'cloud point' temperature does not occur. This is likely a result of the fact that HT and 1,6-HD are weakly surface active, with no apparent critical aggregation concentration (CAC); when including the PDC, the 'cloud point' temperature is similar to that of the surfactant solution alone rather than the mixture. Therefore, they behave more as co-solutes rather than co-surfactants, providing an alternate hydrophobic surface with which the surfactant monomers can interact. Hexane-1,2-diol, which lowers the 'cloud point' temperature and causes a shift to lower ionic strengths for the onset of attractive PDC interactions, has a clear CAC of 0.81 M. Therefore, it is likely that the surface activity of 1,2-HD allows it to partition into the surfactant portion of the PDC, thereby influencing its interactions. Comparing B22 values at the various solution conditions for which a surfactant 'cloud point' temperature was observed, a strong correlation appears. This clearly points to the constructive role surfactant plays in promoting PDC crystallization, and is similar to the type of ordering observed for colloidal particles approaching a liquid - liquid phase boundary in structured surfactant solutions. Although the same range of weakly attractive B22 interactions tends to promote crystallization for both salts and PEG, the corresponding 'cloud point' temperatures are different. This may reflect the different underlying forces involved in phase separation; studies on model soluble proteins found strong correlation between the liquid - liquid and solid - liquid temperatures, with B22 values leading to crystallization all

	falling within a similar range, yet different values approaching the critical point. Likewise, simulations of colloidal particles experiencing short range attraction found a direct relationship between the range of interaction and the reduced critical temperature. Thus in this case, where the cloud point, as a surrogate to the critical temperature, varies based on the precipitants used, the range of interaction, or B22, remains within a certain range for crystallization. It will be interesting to see whether additional membrane proteins and surfactants display similar behavior. Previously, we had performed small-angle neutron scattering (SANS) experiments to investigate the microstructure and interactions in mixtures of the nonionic surfactants C8ßG1 and C8E3 at conditions close to the cloud point, where well-organized crystals of membrane proteins are most often obtained. Results indicated that small globular micelles exist in these mixtures, which maintain size and shape while attractive interactions increase progressively as the system approaches the phase boundary. More recently, SANS experiments were carried out on protein-surfactant complexes of bacteriorhodopsin in C8ßG1/C8E3 mixtures. A model was developed to decouple particle form factor and structure factor of the protein-surfactant complexes, once the microstructure of the empty micelles was characterized. Results show that the interactions between protein-surfactant complexes become progressively more attractive as the surfactant mixture becomes richer in C8E3, thus following the same pattern as the protein-free surfactant solutions, which were also measured. These results prove that for bacteriorhodopsin the major component to the interactions among complexes is due to the surfactant bound to the hydrophobic part of the protein. We also extended the characterization of surfactant solutions for potential use in membrane protein crystallization. In particular, we found interesting patterns of phase behavior for short chain alkyl monoglucoside surfactants (C8ßG1 and C9ßG1) in solutions with polyethylene glycol (PEG) and ammonium sulfate, which are commonly used as precipitants in crystallization screens. These surfactants are soluble in aqueous solutions of ammonium sulfate up to 4 M ionic strength. In combination with PEG of molecular weight from 3350 to 10000, two phase transitions can be observed, with lower and upper two-phase regions separated by a single isotropic solution. The lower miscibility region becomes larger with increasing the PEG molecular weight or the surfactant hydrophobic tail. SANS experiments were used to characterize the microstructure of these solutions in the one-phase regions near the lower miscibility gap. The scattering data were evaluated with the Generalized Indirect Fourier Transformation (GIFT) technique, which allowed to separate information on the size and shape of the glucoside micelles and their interactions in solutions with different ratios of PEG 3350. Results showed that solutions with C8ßG1 near the cloud point are characterized by weakly attractive interactions that fall in the right range for crystallization, thus explaining the reasons of the increasing success of these combinations in crystallizing membrane proteins.

Bibliography Type:	Description: (Last Updated: 03/17/2017)
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